Double integrals:

The definite integral $\int_a^b f(x)dx$ is defined as limit of the sum

$f(x_1)\delta x_1 + f(x_2)\delta x_2 + f(x_3)\delta x_3 + ... + f(x_n)\delta x_n$ where $n \to \infty$ and each of $\delta x_1, \delta x_2, \delta x_3, ..., \delta x_n$ tend to zero. A double integral is counterpart in two dimensions.

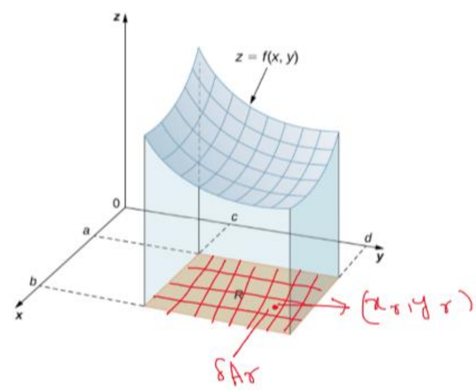
Consider the function $f(x, y)$ of two independent variables x, y defined at each point in the finite region R of xy $-$plane. Divide R into n elementary areas $\delta A_1, \delta A_2, \delta A_3, ..., \delta A_n$. Let (x_r, y_r) be any point within the r^{th} elementary area δA_r. Consider the sum

$f(x_1, y_1)\delta A_1 + f(x_2, y_2)\delta A_2 + f(x_3, y_3)\delta A_3 + ... + f(x_n, y_n)\delta A_n$. i.e., $\sum_{r=1}^{n} f(x_r, y_r)\delta A_r$

The limit of this sum if exists as number of subdivisions increases indefinitely and area of each subdivision decreases to zero, is defined as the double integral of $f(x, y)$ over the region R and is written as

$$\iint_R f(x, y)dA$$

Thus $\iint_R f(x, y)dA = \lim_{\substack{n \to \infty \\ \delta A \to 0}} \sum_{r=1}^{n} f(x_r, y_r)\delta A_r$ --------(1)

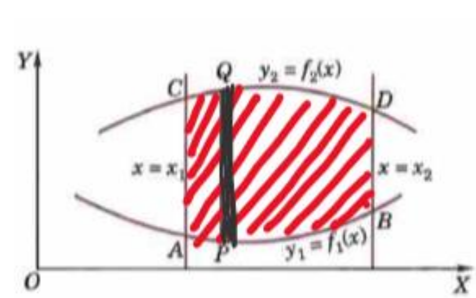


For the purpose of evaluation (1) is expressed as

The repeated integral $\int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x, y)dxdy$

Case 1: When y_1 and y_2 are functions of x and x_1 and x_2 are constants $f(x, y)$ is first integrated with respect to y keeping x fixed between the limits y_1 and y_2 and then resulting expression is integrated with respect to x within the limits x_1 and x_2

i.e., $I_1 = \int_{x_1}^{x_2}\left\{\int_{y_1}^{y_2} f(x, y)dy\right\}dx$

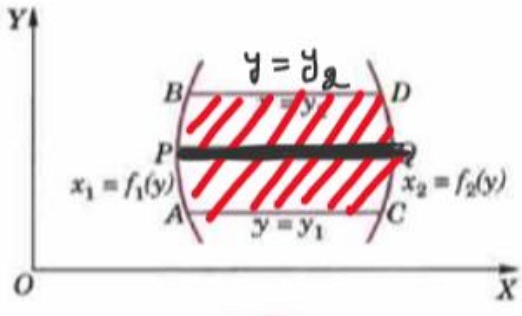


AB and CD are the two curves whose equations are $y_1 = f_1(x)$ and $y_2 = f_2(x)$ respectively. PQ is a vertical strip of width dx. Whole region of integration is $ABCD$, where x value varies from x_1 to x_2 .

Case 2: When x_1 and x_2 are functions of y and y_1 and y_2 are constants $f(x, y)$ is first integrated with respect to x keeping y fixed between the limits x_1 and x_2 and then resulting expression is integrated with respect to y within the limits y_1 and y_2

i.e., $I_1 = \int_{y_1}^{y_2} \left\{ \int_{x_1}^{x_2} f(x,y)dx \right\} dy$

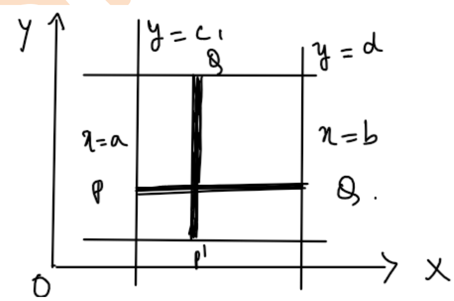
AB and CD are the two curves whose equations are $x_1 = f_1(y)$ and $x_2 = f_2(y)$ respectively. PQ is a horizontal strip of width dy. Whole region of integration is $ABCD$, where y value varies from y_1 to y_2



Case 3: when both pairs of limits are constants, the region of integration is $ABCD$ as shown in the figure below.

Thus, for constant limits it hardly matters whether we first integrate wrt x then wrt y or viceversa.
Therefore,

$$\int_a^b \int_c^d f(x,y)dydx = \int_c^d \int_a^b f(x,y)dxdy$$



Note: If $f(x,y)$ is not continuous at any point in the rectangular region R then $\int_a^b \int_c^d f(x,y)dydx$ need not be equal to $\int_c^d \int_a^b f(x,y)dxdy$

Examples:

1. Evaluate $\int_0^1 \int_0^1 \frac{dxdy}{x+y+1}$

Solution: $\int_0^1 \int_0^1 \frac{dxdy}{x+y+1} = \int_0^1 \left\{ \int_0^1 \frac{dx}{x+y+1} \right\} dy = \int_0^1 \left[\log(x+y+1) \right]_0^1 dy$

$$= \int_0^1 [\log(2+y) - \log(y+1)]dy \quad = \int_0^1 \log(2+y)dy - \int_0^1 \log(y+1)dy$$

$$= \int_0^1 \log(2+y)dy - \int_0^1 \log(y+1)dy \quad = [y\log(2+y)]_0^1 - \int_0^1 \frac{y}{y+2}dy - [y\log(y+1)]_0^1 + \int_0^1 \frac{y}{y+1}dy$$

$$= \log 3 - 0 - \int_0^1 \frac{y+2-2}{y+2}dy - [\log 2 - 0] + \int_0^1 \frac{y+1-1}{y+1}dy \quad = \log\frac{3}{2} - \int_0^1 dy + 2\int_0^1 \frac{1}{y+2}dy + \int_0^1 dy - \int_0^1 \frac{1}{y+1}dy$$

$$= \log\frac{3}{2} + 2[\log(y+2)]_0^1 - [\log(y+1)]_0^1 \quad = \log\frac{3}{2} + 2[\log 3 - \log 2] - [\log 2 - \log 1] = \log\left(\frac{27}{16}\right)$$

2. Evaluate $\int_0^{\pi}\int_0^{\sin y} dxdy$

Solution: $\int_0^{\pi}\int_0^{\sin y} dxdy = \int_0^{\pi}[x]_0^{\sin y} dy = \int_0^{\pi}\sin y dy = [-\cos y]_0^{\pi} = -\cos\pi + \cos 0 = 1 + 1 = 2$

3. Evaluate $\int_0^1\int_x^{\sqrt{x}} \left(x^2+y^2\right)dydx$

Solution: $\int_0^1\int_x^{\sqrt{x}} \left(x^2+y^2\right)dydx = \int_0^1\left[x^2y + \frac{y^3}{3}\right]_x^{\sqrt{x}} dx = \int_0^1\left[x^2\left(\sqrt{x}-x\right) + \frac{\left(\sqrt{x}\right)^3 - x^3}{3}\right]dx$

$$= \int_0^1\left[x^{\frac{5}{2}} - x^3 + \frac{x^{\frac{3}{2}} - x^3}{3}\right]dx = \left[\frac{2}{7}x^{\frac{7}{2}} - \frac{4}{3}\frac{x^4}{4} + \frac{1}{3}\frac{2}{5}x^{\frac{5}{2}}\right]_0^1 = \frac{3}{35}$$

Jacobians: Suppose u and v are functions of two independent variables x and y then the determinant $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ is called Jacobian of u and v with respect x and y and is denoted by $\frac{\partial(u,v)}{\partial(x,y)}$ or $J\left(\frac{u,v}{x,y}\right)$ or J

Also if u, v, w are functions of three independent variables x, y, z then, Jacobian of u , v and w with respect x, y and z is given by,

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

. It is also denoted by $J\left(\frac{u,v,w}{x,y,z}\right)$ or J

Change of variables : Let $x = x(u,v)$ and $y = y(u,v)$ and $\dfrac{\partial(x,y)}{\partial(u,v)} \neq 0$, then $\iint\limits_{R} f(x,y)dxdy = \iint\limits_{\bar{R}} \phi(u,v)J dudv$ where R is the region where x *and* y vary and $\bar{R}$ is the region where u *and* v vary and $\phi(u,v) = f(x(u,v), y(u,v))$.

Double integral in polar form:

Let $x = r\cos\theta, y = r\sin\theta$

Then $\dfrac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial y}{\partial r} \\ \dfrac{\partial x}{\partial \theta} & \dfrac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$

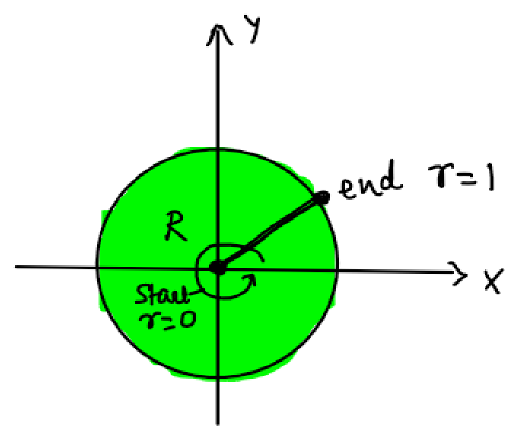
Therefore, $\iint\limits_{R} f(x,y)dxdy = \iint\limits_{\bar{R}} \phi(r,\theta) r dr d\theta$, where R is the region where x *and* y vary and $\bar{R}$ is the region where u *and* v vary.

Examples:

1. Evaluate $\iint\limits_{R} \dfrac{(x-y)^2}{x^2+y^2} dxdy$ by changing to polar coordinates, where R is the region $x^2+y^2 \leq 1$

Solution: Let $x = r\cos\theta, y = r\sin\theta$. Then $dxdy = r dr d\theta$.

In R: $x^2+y^2 \leq 1$, r varies from 0 *to* a and θ varies from 0 *to* 2π.



$$\frac{(x-y)^2}{x^2+y^2} = \frac{r^2(\cos\theta-\sin\theta)^2}{r^2} = (\cos\theta-\sin\theta)^2$$

$$= \cos^2\theta - 2\sin\theta\cos\theta + \sin^2\theta = 1 - 2\sin\theta$$

$$\iint\limits_{R} \frac{(x-y)^2}{x^2+y^2} dxdy = \int_{\theta=0}^{2\pi}\int_{r=0}^{a} (1-2\sin\theta) r dr d\theta$$

$$= \int_{\theta=0}^{2\pi} (1-2\sin\theta) d\theta \int_{r=0}^{a} r dr = [\theta + 2\cos\theta]_0^{2\pi} \left[\frac{r^2}{2}\right]_0^a$$

$$= 2\pi \frac{a^2}{2} = \pi a^2$$

2. Evaluate $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \dfrac{1}{(1+x^2+y^2)^{3/2}} dxdy$ by changing to polar coordinates.

Solution: Let $x = r\cos\theta, y = r\sin\theta$. Then $dxdy = r dr d\theta$.

Region of integration is entire $xy - plane$.

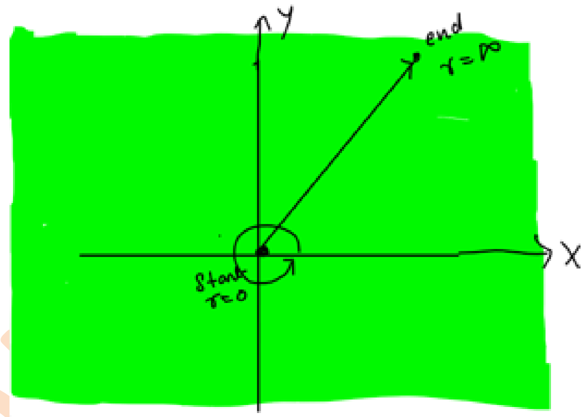
$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{1}{\left(1+x^2+y^2\right)^{3/2}}dxdy=\int_0^{2\pi}\int_0^{\infty}\frac{1}{\left(1+r^2\right)^{3/2}}rdrd\theta \;=\int_0^{2\pi}d\theta\int_0^{\infty}\frac{1}{\left(1+r^2\right)^{3/2}}rdr$$

Let $1+r^2=t\Rightarrow 2rdr=dt\Rightarrow rdr=\frac{dt}{2}$. Also, $r=0\Rightarrow t=1 \, and \, r=\infty\Rightarrow t=\infty$

Therefore,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{1}{\left(1+x^2+y^2\right)^{3/2}}dxdy=[\theta]_0^{2\pi}\int_1^{\infty}\frac{1}{t^{3/2}}\frac{dt}{2}$$

$$=2\pi\frac{1}{2}\left.\frac{t^{-1/2}}{\left(-\frac{1}{2}\right)}\right]_1^{\infty}=-2\pi(0-1)=2\pi$$



Change of order of integration:

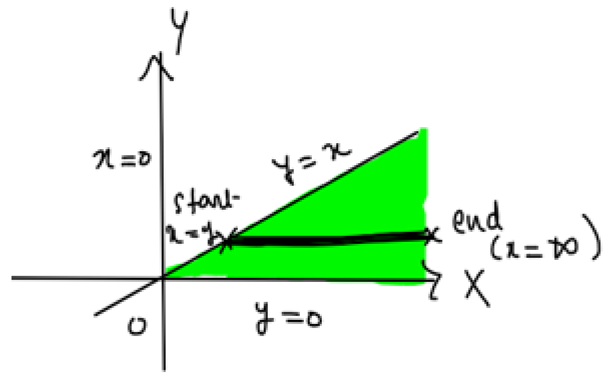
While changing the order of integration we need to convert the integral $\int_{x=a}^{b}\left\{\int_{y=y_1(x)}^{y_2(x)}f(x,y)dy\right\}dx$ into the form $\int_{y=c}^{d}\left\{\int_{x=x_1(y)}^{x_2(y)}f(x,y)dx\right\}dy$ or vice versa. This procedure generally changes the limits of integration and new limits are to be determined by examining the geometrical region in which the integration is being carried out. We use this method because, sometimes the evaluation becomes very difficult, but after changing the order it becomes very easy.

Examples:

1. Evaluate $\int_0^{\infty}\int_0^{x}xe^{-\frac{x^2}{y}}dydx$ by changing the order of integration.

Solution: Here y varies from $y=0 \; to \; y=x$ and x varies from $0 \; to \; \infty$. The region of integration is as shown in the figure.

While changing the order of integration we need to integrate with respect to x and then with respect to y. So, let us consider a horizontal strip (strip parallel to x axis) in the region of integration, whose starting point lies on the line $x = y$ and end point lies at ∞. The least and greatest value of y in the region are 0 *and* ∞ respectively.



Therefore,

$$\int_0^\infty\int_0^x xe^{-\frac{x^2}{y}}dydx = \int_y^\infty\int_0^\infty xe^{-\frac{x^2}{y}}dxdy = \int_y^\infty\left\{\int_0^\infty xe^{-\frac{x^2}{y}}dx\right\}dy$$

Let $\frac{x^2}{y} = t \Rightarrow x^2 = yt \Rightarrow xdx = \frac{ydt}{2}$

$x = 0 \Rightarrow t = 0$ *and* $x = \infty \Rightarrow t = \infty$

Thus,

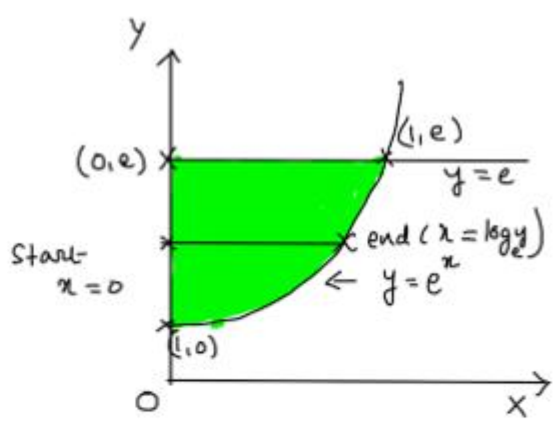
$$\int_0^\infty\int_0^x xe^{-\frac{x^2}{y}}dydx = \frac{1}{2}\int_0^\infty\left\{\int_y^\infty ye^{-t}dt\right\}dy = \frac{1}{2}\int_0^\infty\left\{-y\left(e^{-t}\right)_y^\infty\right\}dy$$

$$= \frac{1}{2}\int_0^\infty ye^{-y}dy = \frac{1}{2}\left[-ye^{-y} - e^{-y}\right]_0^\infty = \frac{1}{2}$$

2. Evaluate $\int_0^1\int_{e^x}^e \frac{dydx}{\log y}$ by changing the order of integration.

Solution: Here y varies from $y = e^x$ *to* $y = e$ and x varies from 0 *to* 1. The region of integration is as shown in the figure.

While changing the order of integration we need to integrate with respect to x first and then with respect to y. So, let us consider a horizontal strip (strip parallel to x axis) in the region of integration, whose starting point lies on the line $x = 0$ and end point lies at $x = \ln y$. The least and greatest value of y in the region are 1 *and* e respectively.



Therefore,

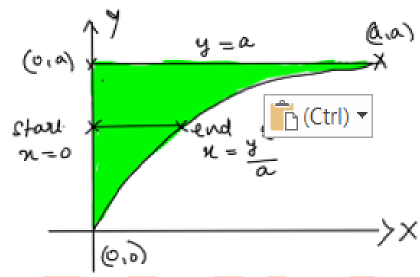
$$\int_0^1\int_{e^x}^e \frac{dydx}{\log y} = \int_1^e\left\{\int_0^{\log y}\frac{dx}{\log y}\right\}dy = \int_1^e \frac{1}{\log y}\{x\}_0^{\log y}dy$$

$$= \int_1^e \frac{1}{\log y}\{\log y\}dy = (y)_1^e = e - 1$$

3. Change the order of integration and hence evaluate, $I=\int_0^a\int_{\sqrt{ax}}^{a}\frac{y^2dxdy}{\sqrt{y^4-a^2x^2}}$

Solution: Given that y varies from $y=\sqrt{ax}$ *to* $y=a$ and x varies from $x=0$ *to* $x=a$
The region of integration is as shown in the figure.

While changing the order of integration we need to integrate with respect to x first and then with respect to y. So, let us consider a horizontal strip (strip parallel to x axis) in the region of integration, whose starting point lies on the line $x=0$ and end point lies at $x=\frac{y^2}{a}$. The least and greatest value of y in the region are 0 *and* a respectively.



Therefore, $I=\int_0^a\int_{\sqrt{ax}}^{a}\frac{y^2dxdy}{\sqrt{y^4-a^2x^2}}=\int_0^a\left\{\int_0^{\frac{y^2}{a}}\frac{y^2dx}{\sqrt{y^4-a^2x^2}}\right\}dy=\int_0^a\left\{\int_0^{\frac{y^2}{a}}\frac{y^2dx}{y^2\sqrt{1-a^2\frac{x^2}{y^4}}}\right\}dy$

Let $\frac{ax}{y^2}=\sin\theta\Rightarrow x=\frac{y^2\sin\theta}{a}\Rightarrow dx=\frac{y^2\cos\theta d\theta}{a}$

$x=0\Rightarrow\theta=0$ *and* $x=\frac{y^2}{a}\Rightarrow\theta=\frac{\pi}{2}$

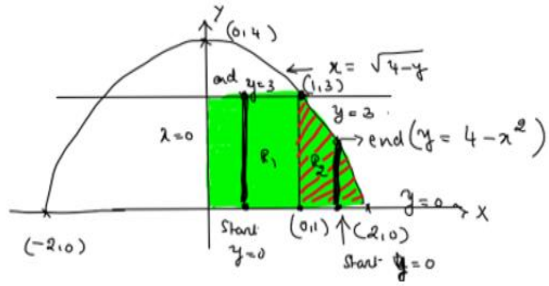
Thus, $I=\int_0^a\left\{\int_0^{\frac{\pi}{2}}\frac{y^4\cos\theta d\theta}{ay^2\sqrt{1-\sin^2\theta}}\right\}dy=\int_0^a\frac{y^2}{a}\left\{\int_0^{\frac{\pi}{2}}d\theta\right\}dy=\int_0^a\frac{\pi}{2a}y^2dy$

$$=\frac{\pi}{2a}\left[\frac{y^3}{3}\right]_0^a=\frac{\pi a^2}{6}$$

4. Change the order of integration in $I=\int_0^3\int_0^{\sqrt{4-y}}(x+y)dxdy$ and hence evaluate it.

Solution: Given that x varies from $x=0$ *to* $x=\sqrt{4-y}$ and y varies from 0 *to* 3.The region of integration is as shown in the figure.

While changing the order of integration we need to integrate with respect to y first and then with respect to x. So, let us consider a vertical strip (strip parallel to y axis) in the region of integration. The region of integration is divided into two subdivisions R_1 *and* R_2 . in R_1,y varies from 0 *to* 3.and x varies from 0 *to* 1. In R_2, y varies from $y = 0$ *to* $y = 4 - x^2$ and x varies from 1 *to* 2.



Therefore,

$$I = \int_{x=0}^{1}\int_{y=0}^{3}(x+y)dydx + \int_{x=1}^{2}\int_{y=0}^{4-x^2}(x+y)dydx$$

$$= \int_{x=0}^{1}(xy+\frac{y^2}{2})_0^3 dx + \int_{x=1}^{2}(xy+\frac{y^2}{2})_0^{4-x^2}dx = \int_{x=0}^{1}(3x+\frac{9}{2})dx + \int_{x=1}^{2}(x(4-x^2)+\frac{(4-x^2)^2}{2})dx$$

$$= \left(3\frac{x^2}{2}+\frac{9}{2}x\right)_0^1 + \left(4\frac{x^2}{2}-\frac{x^4}{4}\right)_1^2 + \left(\frac{16}{2}x-\frac{8x^3}{6}+\frac{x^5}{10}\right)_1^2$$

$$= \left(\frac{3}{2}+\frac{9}{2}\right)+\left(\frac{4\times3}{2}-\frac{15}{4}\right)+\left(\frac{16}{2}-\frac{8\times7}{6}+\frac{31}{10}\right)$$

$$= 6+6-\frac{15}{4}+8-\frac{28}{3}+\frac{31}{10}=\frac{601}{60}$$

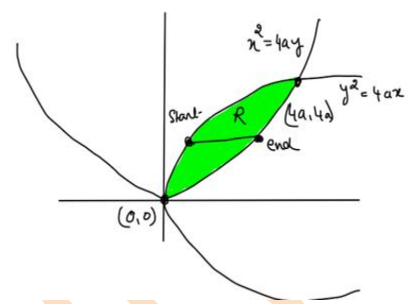
Area, volume and average value of a function:

1. $\iint_R zdxdy$ gives the volume of solid bounded by the region R and the surface $z = f(x,y)$ where R is projection of the surface $z = f(x,y)$ on $xy - plane$. i.e., $\iint_R zdxdy$ gives volume under the surface $z = f(x,y)$ above $xy - plane$.
2. $\iint_R dxdy$ gives area of the region R in cartesian form.
3. $\iint_R rdrd\theta$ gives area of the region R in polar form.
4. $\dfrac{\iint_R f(x,y)dxdy}{\iint_R dxdy}$ or $\dfrac{\iint_R f(x,y)dxdy}{A(R)}$ gives the average value of the function $f(x,y)$ over the region R where $A(R)$ is area of the region R. average value gives the measure of average height of calculated volume.

Examples:

1. Show that the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\dfrac{16a^2}{3}$

Solution:
To find the point of intersection of the parabolas,
$y^2 = 4ax$ -----(1) and $x^2 = 4ay$ ----(2):
By (1) and (2),



$$\left(\frac{y^2}{4a}\right)^2 = 4ay \Rightarrow y(y^3 - (4a)^3) = 0 \Rightarrow y = 4a \ or \ y = 0$$

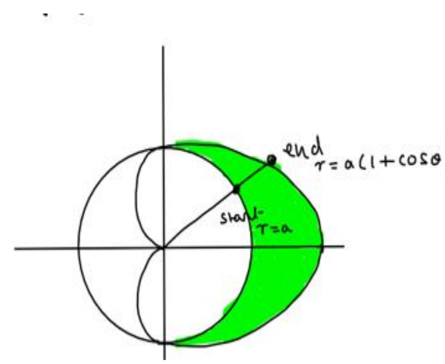
$$\Rightarrow x = 4a \ or \ x = 0$$

Therefore, the points of intersection are,
$(0,0)$ *and* $(4a,4a)$
Area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is given by,

$$\iint_R dxdy = \int_{y=0}^{4a} \left\{ \int_{x=\frac{y^2}{4a}}^{2\sqrt{ay}} dx \right\} dy = \int_{y=0}^{4a} \left\{ 2\sqrt{ay} - \frac{y^2}{4a} \right\} dy$$

$$= \left[2\sqrt{a}\,\frac{y^{3/2}}{(3/2)} - \frac{1}{4a}\frac{y^3}{3} \right]_0^{4a} = 2\sqrt{a}\,\frac{(4a)^{3/2}}{(3/2)} - \frac{1}{4a}\frac{(4a)^3}{3} = \frac{16a^2}{3}$$

2. Find the area lying inside the cardioid $r = a(1 + cos\theta)$ and outside the circle $r = a$



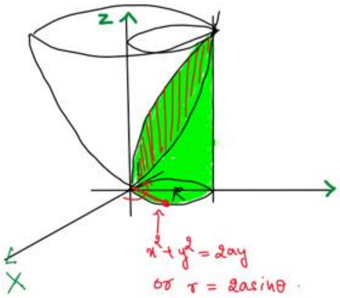
Solution: Required are= $2\times$area above $x - axis$)

$$2\int_0^{\frac{\pi}{2}} \int_a^{a(1+\cos\theta)} r dr d\theta = 2\int_0^{\frac{\pi}{2}} \left[\frac{r^2}{2}\right]_a^{a(1+\cos\theta)} d\theta = a^2 \int_0^{\frac{\pi}{2}} \left[(1+\cos\theta)^2 - 1\right] d\theta$$

$$= a^2 \int_0^{\frac{\pi}{2}} \left[2\cos\theta + \cos^2\theta\right] d\theta = a^2 \int_0^{\frac{\pi}{2}} \left[2\cos\theta + \frac{1+\cos 2\theta}{2}\right] d\theta$$

$$= a^2 \left[2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}} = a^2\left[2 + \frac{\pi}{4}\right]$$

3. Find the volume bounded by the paraboloid $x^2+y^2=az$, the cylinder $x^2+y^2=2ay$ and the plane $z=0$



Solution: the region of integration is as shown in the diagram.

Required volume $=\iint_R z\,dx\,dy$ where $z=\dfrac{x^2+y^2}{a}$ and R is the circle $x^2+y^2=2ay$

Let $x=r\cos\theta, y=r\sin\theta$. Then $dxdy = r\,dr\,d\theta$.

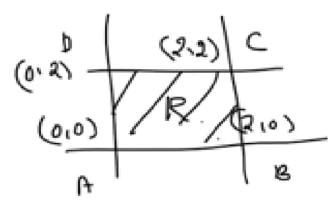
$x^2+y^2=2ay \Rightarrow r=2a\sin\theta$ and $z=\dfrac{x^2+y^2}{a}\Rightarrow z=\dfrac{r^2}{a}$.

Therefore, required volume $=\displaystyle\int_{\theta=0}^{\pi}\int_{r=0}^{2a\sin\theta}\frac{r^2}{a}r\,dr\,d\theta=\frac{1}{a}\int_{\theta=0}^{\pi}\int_{r=0}^{2a\sin\theta}r^3\,dr\,d\theta$

$$=\frac{1}{a}\int_{\theta=0}^{\pi}\left[\frac{r^4}{4}\right]_{r=0}^{2a\sin\theta}d\theta=\frac{1}{4a}\int_{\theta=0}^{\pi}(2a)^4\sin^4\theta\,d\theta=4a^3\times 2\int_{\theta=0}^{\frac{\pi}{2}}\sin^4\theta\,d\theta$$

$$=8a^3\,\frac{3}{4}\,\frac{1}{2}\,\frac{\pi}{2}=\frac{3}{2}\pi a^3$$

4. Determine the average value of $f(x,y)=e^{x+y}$ over the region $R=[0,2]\times[0,\ 2]$



Solution: we know that average value of $f(x,y)$ over the region $R=\dfrac{\iint_R f(x,y)dxdy}{area\,of\ R}$.

Therefore, average value of e^{x+y} over the region $[0,2]\times[0,2]$

$$=\frac{\int_0^2\int_0^2 e^{x+y}dxdy}{Area\,of\ ABCD}=\frac{1}{2^2}\int_0^2 e^x dx\int_0^2 e^y dy=\frac{1}{4}\left(e^2\right)_0^2\left(e^2\right)_0^2=\frac{e^2-1}{4}$$

Triple integrals: Consider a function $\boldsymbol{f(x,y,z)}$ defined at every point of three-dimensional finite region V. Divide V into elementary volumes $\delta V_1,\delta V_2,\delta V_{3,\ldots\ldots,}\delta V_n$. Let (x_r,y_r,z_r) be any point within the r^{th} subdivision δV_r. Consider the sum, $\sum_{r=1}^{\infty} f(x_r,y_r,z_r)\delta V_r$. The limit of the sum, if exists, as $n\to\infty$, $\delta V_r\to 0$ is called triple integral of $\boldsymbol{f(x,y,z)}$ over the region V and is denoted by $\iiint_V f(x,y,z)dV$.

For the purpose of evaluation, it can be expressed as repeated integral $\int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} f(x,y,z)dzdydx$. If $x_1 and\ x_2$ are constants, $y_1 and\ y_2$ are constants or functions of x, $z_1 and\ z_2$ are constants or functions of x,y then the integral is evaluated as follows:

First $f(x,y,z)$ is integrated with respect to z between the limits $z_1 and\ z_2$ keeping $x\ and\ y$ fixed. The resulting expression is integrated with respect to y between the limits $y_1 and\ y_2$ keeping x constant. The result obtained is finally integrated with respect to x between the limits $x_1 and\ x_2$.

Thus, $I=\int_{x_1}^{x_2}\left\{\int_{y_1}^{y_2}\left\{\int_{z_1}^{z_2} f(x,y,z)dz\right\}dy\right\}dx$.

Examples:

1. Evaluate $\int_0^1\int_0^2\int_1^2 xyz^2dxdydz$

Solution: $\int_0^1\int_0^2\int_1^2 xyz^2dxdydz=\int_0^1 z^2dz\int_0^2 ydy\int_1^2 xdx=\left[\frac{z^3}{3}\right]_0^1\left[\frac{y^2}{2}\right]_0^2\left[\frac{x^2}{2}\right]_1^2$

$=\frac{1}{3}\frac{4}{2}\left[\frac{4-1}{2}\right]=1$

2. Evaluate $\int_0^1\int_{y^2}^1\int_0^{1-x} xdzdxdy$

Solution: $\int_0^1\int_{y^2}^1\int_0^{1-x} xdzdxdy=\int_0^1\int_{y^2}^1 x[z]_0^{1-x}dxdy=\int_0^1\int_{y^2}^1 x[1-x]dxdy=\int_0^1\int_{y^2}^1\left[x-x^2\right]dxdy$

$$=\int_0^1\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_{y^2}^1dy=\int_0^1\left[\frac{1}{2}-\frac{1}{3}-\frac{y^4}{2}+\frac{y^6}{3}\right]dy=\left[\frac{1}{6}y-\frac{y^5}{10}+\frac{y^7}{21}\right]_0^1$$

$$=\frac{1}{6}-\frac{1}{10}+\frac{1}{21}=\frac{35-21+10}{210}=\frac{24}{210}=\frac{4}{35}$$

3. Evaluate $\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} xyzdzdxdy$

Solution: $\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} xyzdzdxdy=\int_0^1\int_0^{\sqrt{1-x^2}}\left(\int_0^{\sqrt{1-x^2-y^2}} zdz\right)xydxdy=\int_0^1\int_0^{\sqrt{1-x^2}}\left(\frac{z^2}{2}\right)_0^{\sqrt{1-x^2-y^2}} xydxdy$

$=\int_0^1\int_0^{\sqrt{1-x^2}}\left(\frac{1-x^2-y^2}{2}\right)xydydx=\frac{1}{2}\int_0^1\int_0^{\sqrt{1-x^2}}\left[\left(x-x^3\right)y-xy^3\right]dydx=\frac{1}{2}\int_0^1\left[\left(x-x^3\right)\frac{y^2}{2}-x\frac{y^4}{4}\right]_0^{\sqrt{1-x^2}}dx$

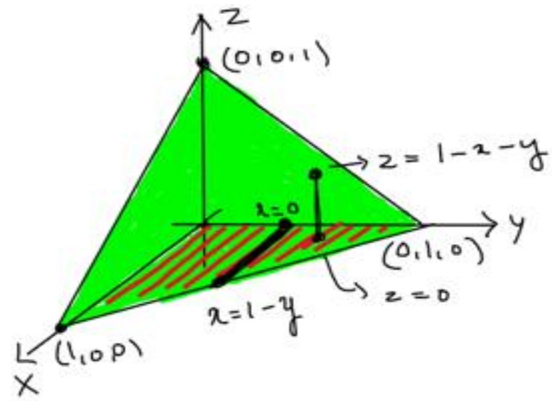
$=\frac{1}{2}\int_0^1\left[\left(x-x^3\right)\frac{\left(1-x^2\right)}{2}-x\frac{\left(1-x^2\right)^2}{4}\right]dx=\frac{1}{8}\int_0^1\left[x\left(1-x^2\right)^2\right]dx=\frac{1}{8}\int_0^1\left[x\left(1-2x^2+x^4\right)\right]dx$

$=\frac{1}{8}\int_0^1\left[\left(x-2x^3+x^5\right)\right]dx=\frac{1}{8}\left[\frac{x^2}{2}-2\frac{x^4}{4}+\frac{x^6}{6}\right]_0^1=\frac{1}{8}\times\frac{1}{6}=\frac{1}{48}$

4. Evaluate $\iiint(x+y+z)dxdydz$ over the tetrahedron bounded by the planes $x=0, y=0, z=0$ and $x+y+z=1$.

Solution: The region of integration is as shown in the diagram where z varies from XY plne $(z = 0)$ to the plane $x + y + z = 1$ i.e., $z = 1 - x - y$. x varies from $x = 0$ to the line $x + y = 1$
i.e., $x = 1 - y$ and y varies from 0 to 1.

Therefore, $\iiint (x+y+z)dxdydz = \int_0^1\int_0^{1-y}\int_0^{1-x-y}(x+y+z)dxdydz$



$$= \int_0^1\int_0^{1-y}\left[\int_0^{1-x-y}(x+y+z)dz\right]dxdy = \int_0^1\int_0^{1-y}\left[(x+y)z+\frac{z^2}{2}\right]_0^{1-x-y}dxdy$$

$$= \int_0^1\int_0^{1-y}\left[(x+y)(1-x-y)+\frac{(1-x-y)^2}{2}\right]dxdy$$

$$= \int_0^1\int_0^{1-y}\left[(1-x-y)\left[(x+y)+\frac{(1-x-y)}{2}\right]\right]dxdy$$

$$= \frac{1}{2}\int_0^1\int_0^{1-y}\left[(1-x-y)[1+x+y]\right]dxdy = \frac{1}{2}\int_0^1\int_0^{1-y}\left[\left(1-(x+y)^2\right)\right]dxdy = \frac{1}{2}\int_0^1\left[x-\frac{(x+y)^3}{3}\right]_0^{1-y}dy$$

$$= \frac{1}{2}\int_0^1\left[(1-y)-\frac{(1-y+y)^3}{3}+\frac{y^3}{3}\right]dy = \frac{1}{2}\int_0^1\left[(1-y)-\frac{1}{3}+\frac{y^3}{3}\right]dy = \frac{1}{2}\left[y-\frac{y^2}{2}-\frac{y}{3}+\frac{y^4}{12}\right]_0^1$$

$$= \frac{1}{2}\left[1-\frac{1}{2}-\frac{1}{3}+\frac{1}{12}\right] = \frac{1}{8}$$

Change of variables:

We can often reduce computational work while evaluating triple integrals by changing variables x, y, z to new variables u, v, w such that,

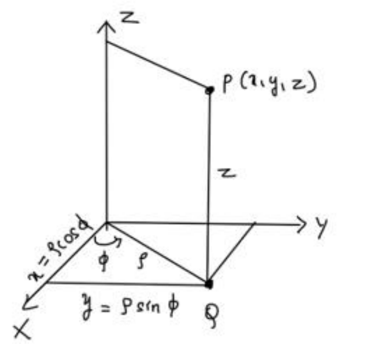
$J = \dfrac{\partial(x,y,z)}{\partial(u,v,w)} \neq 0$.

It can be proved that $\iiint_R f(x,y,z)dxdydz = \iiint_{\bar{R}} \phi(u,v,w)Jdudvdw$, where R is the region in which x, y, z vary

and $\bar{R}$ is the region in which u, v, w vary and $\phi(u,v,w) = f(x(u,v,w), y(u,v,w), z(u,v,w))$

two standard examples of (u, v, w) which are widely used in the evaluation of triple integrals are cylindrical polar coordinates (ρ, ϕ, z) and spherical coordinates (r, θ, ϕ).

Triple integrals in polar co-ordinates:



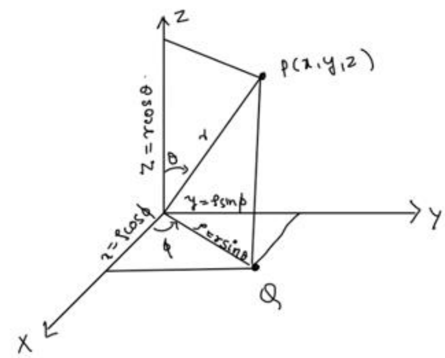
Suppose (x, y, z) are related to three variables (ρ,ϕ,z) through $x=\rho\cos\phi,\ y=\rho\sin\phi, z=z$ then (ρ,ϕ,z) are called cylindrical polar coordinates.

$$J=\frac{\partial(x,y,z)}{\partial(\rho,\phi,z)}=\rho$$

Therefore, $\iiint\limits_R f(x,y,z)dxdydz=\iiint\limits_R \phi(\rho,\phi,z)\rho d\rho d\phi dz$

Triple integrals in cylindrical polar coordinates:

Suppose (x, y, z) are related to three variables (r,θ,ϕ) through $x=r\sin\theta\cos\phi,\ y=r\sin\theta\sin\phi, z=r\cos\theta$ then (r,θ,ϕ) are called spherical polar coordinates.



$$J=\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}=r^2\sin\theta$$

Therefore, $\iiint\limits_R f(x,y,z)dxdydz=\iiint\limits_R \phi(r,\theta,\phi)r^2\sin\theta drd\theta d\phi$, where $r\geq 0, 0\leq\theta\leq\pi, 0\leq\phi\leq 2\pi$

Examples:

1. Find by triple integration volume of the sphere $x^2+y^2+z^2=a^2$

Solution: Changing to spherical coordinates, by putting $x=r\sin\theta\cos\phi,\ y=r\sin\theta\sin\phi, z=r\cos\theta$ .Then, $dxdydz = r^2\sin\theta drd\theta d\phi$

Area of sphere= 8 ×Volume of its portion in first octant.

$$=8\int_{\phi=0}^{\frac{\pi}{2}}\int_{\theta=0}^{\frac{\pi}{2}}\int_{r=0}^{a} r^2\sin\theta drd\theta d\phi=8\int_{\phi=0}^{\frac{\pi}{2}} d\phi\int_{\theta=0}^{\frac{\pi}{2}}\sin\theta d\theta\int_{r=0}^{a} r^2dr=8[\phi]_0^{\frac{\pi}{2}}[-\cos\theta]_0^{\frac{\pi}{2}}\left[\frac{r^3}{3}\right]_0^a=8\times\frac{\pi}{2}\times 1\times\frac{a^3}{3}=\frac{4\pi a^3}{3}$$

2. Find the volume of the portion of the sphere $x^2+y^2+z^2=a^2$ lying inside the cylinder $x^2+y^2=ay$

Solution: let us use cylindrical co-ordinates, $x=\rho\cos\phi,\ y=\rho\sin\phi, z=z$. Then $dxdydzy=\rho d\rho d\phi dz$.

Equation of the sphere is $x^2+y^2+z^2=a^2$ i.e., $\rho^2+z^2=a^2\Rightarrow z^2=a^2-\rho^2\Rightarrow z=\sqrt{a^2-\rho^2}$

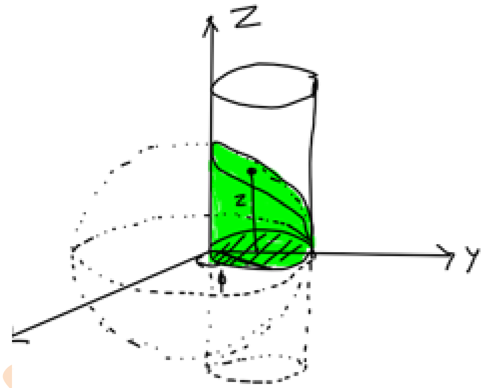
In the region of integration,

z varies from 0 to $\sqrt{a^2-\rho^2}$

ρ varies from 0 to a.

ϕ varies from 0 to π.

Therefore, required volume= $2\times Volume\ above\ XY\ plane$.



$$=2\int_{\phi=0}^{\pi}\int_{\rho=0}^{a\sin\phi}\sqrt{a^2-\rho^2}\int_{z=0}^{\sqrt{a^2-\rho^2}}\rho d\rho d\phi dz=2\int_{\phi=0}^{\pi}\int_{\rho=0}^{a\sin\phi}[z]_0^{\sqrt{a^2-\rho^2}}\rho d\rho d\phi$$

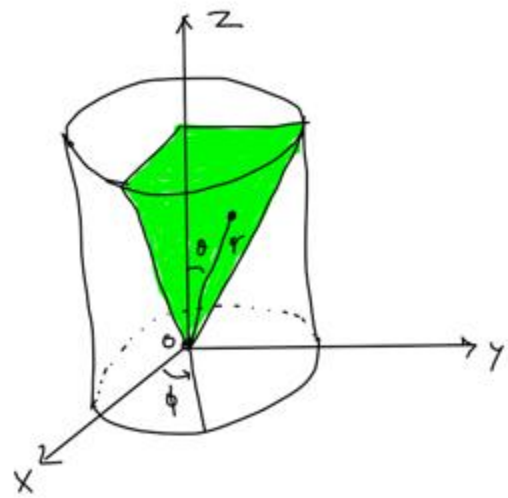
$$=2\int_{\phi=0}^{\pi}\int_{\rho=0}^{a\sin\phi}\sqrt{a^2-\rho^2}\rho d\rho d\phi=-\int_{\phi=0}^{\pi}\frac{\left[\left(a^2-\rho^2\right)^{3/2}\right]_0^{a\sin\phi}}{\frac{3}{2}}d\phi$$

$$=-2\int_{\phi=0}^{\pi}\frac{\left(a^2-a^2\sin^2\phi\right)^{3/2}-\left(a^2\right)^{3/2}}{3}d\phi=-2a^3\int_{\phi=0}^{\pi}\frac{\left(\cos^3\phi\right)-1}{3}d\phi=\frac{2a^3}{3}\int_{\phi=0}^{\pi}\left(1-\cos^3\phi\right)d\phi=\frac{2a^3}{9}(3\pi-4)$$

3. Evaluate $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^{1}\frac{dzdydx}{\sqrt{x^2+y^2+z^2}}$

Solution: Let us use spherical coordinates, $x=r\sin\theta\cos\phi$, $y=r\sin\theta\sin\phi$, $z=r\cos\theta$, then, $dxdydz=r^2\sin\theta drd\theta d\phi$.

The region of integration common to the cone, $z^2=x^2+y^2$ and the cylinder $x^2+y^2=1$ bounded by the plane $z=1$ in the first octant. Therefore, given integral becomes,



$$=\int_{\phi=0}^{\frac{\pi}{2}}\int_{\theta=0}^{\frac{\pi}{4}}\int_{r=0}^{\sec\theta}\frac{1}{r}r^2\sin\theta drd\theta d\phi=\int_{\phi=0}^{\frac{\pi}{2}}\int_{\theta=0}^{\frac{\pi}{4}}\left[\frac{r^2}{2}\right]_0^{\sec\theta}\sin\theta d\theta d\phi$$

$$=\frac{1}{2}\int_{\phi=0}^{\frac{\pi}{2}}\int_{\theta=0}^{\frac{\pi}{4}}\sec^2\theta\sin\theta d\theta d\phi=\frac{1}{2}\int_{\phi=0}^{\frac{\pi}{2}}\int_{\theta=0}^{\frac{\pi}{4}}\sec\theta\tan\theta d\theta d\phi$$

$$=\frac{1}{2}\int_{\phi=0}^{\frac{\pi}{2}}[\sec\theta]_0^{\frac{\pi}{4}}d\phi=\frac{1}{2}\int_{\phi=0}^{\frac{\pi}{2}}\left[\sec\frac{\pi}{4}-\sec 0\right]d\phi$$

$$=\frac{1}{2}\int_{\phi=0}^{\frac{\pi}{2}}\left[\sqrt{2}-1\right]d\phi=\frac{\sqrt{2}-1}{2}[\phi]_0^{\frac{\pi}{2}}=\frac{\sqrt{2}-1}{2}\times\frac{\pi}{2}=\frac{\left(\sqrt{2}-1\right)\pi}{4}$$

Mass, Centre of gravity and Moment of inertia:

Let $\rho=f(x,y)>0$ be surface density (mass/unit area) of a given plane region. The amount of mass M contained in the plane region D is given by, $M=\iint_D\rho dxdy=\iint_D f(x,y)dxdy$.

Let $\rho = f(x,y,z) > 0$ be volume density (mass/unit volume) of a given solid. The amount of mass M contained in the volume D is given by, $M = \iiint_V \rho dxdydz = \iiint_V f(x,y,z)dxdydz$.

Moment of inertia:
Moment of inertia of a plane region D with surface density $\rho = f(x,y)$ relative to X −axis, Y-axis and origin are respectively given by,

$$I_x = \iint_D \rho y^2 dxdy = \iint_D y^2 f(x,y)dxdy$$

$$I_y = \iint_D \rho x^2 dxdy = \iint_D x^2 f(x,y)dxdy$$

$$I_o = I_x + I_y = \iint_D \rho(x^2+y^2)dxdy = \iint_D (x^2+y^2) f(x,y)dxdy$$

Moment of inertia of a solid relative to X −axis, Y-axis and Z-axis are respectively given by,

$$I_x = \iiint_V \rho\left(y^2+z^2\right)dxdydz = \iiint_V \left(y^2+z^2\right) f(x,y,z)dxdydz$$

$$I_y = \iiint_V \rho\left(x^2+z^2\right)dxdydz = \iiint_V \left(x^2+z^2\right) f(x,y,z)dxdydz$$

$$I_z = \iiint_V \rho\left(x^2+y^2\right)dxdydz = \iiint_V \left(x^2+y^2\right) f(x,y,z)dxdydz$$

Centre of gravity/ centre of mass:
The coordinates $(\bar{x},\bar{y})$ of centre of gravity of a plane region D with surface density $\rho = f(x,y)$ containing mass M are,

$$\bar{x} = \frac{\iint_D x\rho dxdy}{M}, \quad \bar{y} = \frac{\iint_D y\rho dxdy}{M}$$

The coordinates $(\bar{x},\bar{y},\bar{z})$ of centre of gravity of a solid volume V with surface density $\rho = f(x,y,z)$ containing mass M are,

$$\bar{x} = \frac{\iiint_V x\rho dxdydz}{M}, \quad \bar{y} = \frac{\iiint_V y\rho dxdydz}{M}, \quad \bar{z} = \frac{\iiint_V z\rho dxdydz}{M}$$

Problems:

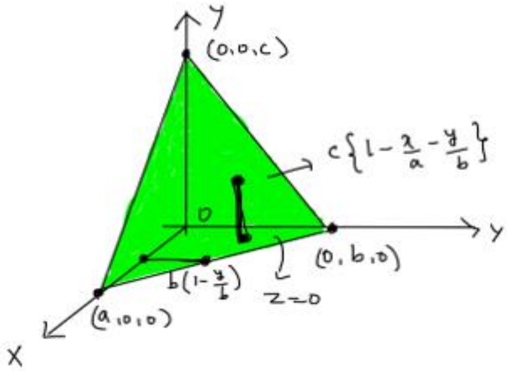
1. Find the mass of the tetrahedron bounded by the coordinate plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c} = 1$, the variable density $\rho = \mu xyz$

Solution: Mass $M = \iiint_V \rho dxdydz = \iiint_V \mu xyz dxdydz$

In the volume V, z varies from XY plne $(z = 0)$ to the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c} = 1$ i.e., $z = c(1-\frac{x}{a}+\frac{y}{b})$.
y varies from $y = 0$ to the line $\frac{x}{a}+\frac{y}{b} = 1$ i.e., $y = b(1-\frac{y}{b})$ and x varies from 0 to a.

Therefore, $M = \iiint_V \rho dxdydz = \iiint_V \mu xyz dxdydz = \int_{x=0}^{a}\int_{y=0}^{b(1-\frac{x}{a})}\int_{z=0}^{c(1-\frac{x}{a}-\frac{y}{b})} \mu xyz dxdydz$

$$= \int_{x=0}^{a}\int_{y=0}^{b(1-\frac{x}{a})} \mu xy\left[\frac{z^2}{2}\right]_0^{c(1-\frac{x}{a}-\frac{y}{b})} dydx = \frac{1}{2}\int_{x=0}^{a}\int_{y=0}^{b(1-\frac{x}{a})} \mu xy\left[c^2(1-\frac{x}{a}-\frac{y}{b})^2\right]dydx$$



$$= \frac{\mu c^2}{2}\int_{x=0}^{a}\int_{y=0}^{b(1-\frac{x}{a})} xy\left[\left(\left(1-\frac{x}{a}\right)-\frac{y}{b}\right)^2\right]dydx$$

$$= \frac{\mu c^2}{2}\int_{x=0}^{a}\int_{y=0}^{b(1-\frac{x}{a})} xy\left(\left(1-\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2-2\left(1-\frac{x}{a}\right)\frac{y}{b}\right)dydx$$

$$= \frac{\mu c^2}{2}\int_{x=0}^{a} x\left(\left(1-\frac{x}{a}\right)^2\left[\frac{y^2}{2}\right]_0^{b(1-\frac{x}{a})}+\frac{x}{b^2}\left[\frac{y^4}{4}\right]_0^{b(1-\frac{x}{a})}-2x\left(1-\frac{x}{a}\right)\left[\frac{y^3}{3b}\right]_0^{b(1-\frac{x}{a})}\right)dx$$

$$= \frac{\mu c^2}{2}\int_{x=0}^{a}\left(x\left\{\frac{b^2}{2}\left(1-\frac{x}{a}\right)^4\right\}+\frac{x}{b^2}\left[\frac{b^4}{4}\right]\left(1-\frac{x}{a}\right)^4-2x\left(1-\frac{x}{a}\right)^4\left[\frac{b^3}{3b}\right]\right)dx$$

$$= \frac{\mu c^2}{2}\int_{x=0}^{a}\left(\frac{b^2}{2}\left\{x\left(1-\frac{x}{a}\right)^4\right\}+\frac{b^2}{4}x\left(1-\frac{x}{a}\right)^4-\frac{2b^2}{3}x\left(1-\frac{x}{a}\right)^4\right)dx = \frac{\mu c^2}{2}\int_{x=0}^{a} xb^2\left(1-\frac{x}{a}\right)^4\left(\frac{1}{2}+\frac{1}{4}-\frac{2}{3}\right)dx$$

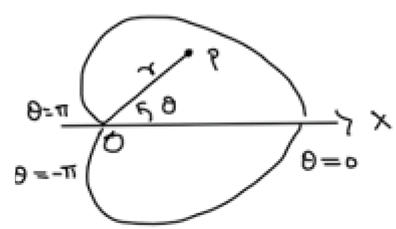
$$= \frac{\mu c^2}{24}\int_{x=0}^{a} xb^2\left(1-\frac{x}{a}\right)^4 dx = \frac{\mu c^2 b^2}{24}\int_{x=0}^{a} x\left(1-\frac{x}{a}\right)^4 dx = \frac{\mu a^2 b^2 c^2}{720}$$

2. Find by double integration the centre of gravity of cardioid $r = a(1+\cos\theta)$

Solution: the cardioid being symmetrical about the initial line, its centre of gravity lies on OX. i.e.,$\bar{y} = 0$.

Therefore,

$$\bar{x} = \frac{\iint_R \rho r dr d\theta}{\iint_R \rho dr d\theta} = \frac{\int_{-\pi}^{\pi}\int_{0}^{a(1+\cos\theta)} \cos\theta r^2 dr d\theta}{\int_{-\pi}^{\pi}\int_{0}^{a(1+\cos\theta)} r dr d\theta} = \frac{\int_{-\pi}^{\pi}\cos\theta\left[\frac{r^3}{3}\right]_0^{a(1+\cos\theta)} d\theta}{\int_{-\pi}^{\pi}\left[\frac{r^2}{2}\right]_0^{a(1+\cos\theta)} d\theta}$$



$$= \frac{\int_{-\pi}^{\pi}\cos\theta\left[\frac{a^3(1+\cos\theta)^3}{3}\right]d\theta}{\int_{-\pi}^{\pi}\left[\frac{a^2(1+\cos\theta)^2}{2}\right]d\theta} = \frac{\frac{a^3}{3}\int_{-\pi}^{\pi}\cos\theta(1+\cos\theta)^3 d\theta}{\frac{a^2}{2}\int_{-\pi}^{\pi}(1+\cos\theta)^2 d\theta}$$

$$= \frac{2a}{3}\frac{\int_{-\pi}^{\pi}\left(\cos\theta+3\cos^2\theta+\cos^4\theta+3\cos^3\theta\right)d\theta}{\int_{-\pi}^{\pi}(1+\cos\theta+\cos^2\theta)d\theta} = \frac{2a}{3}\frac{\int_{-\pi}^{\pi}\left((\cos\theta+3\cos^3\theta)d\theta+\int_{-\pi}^{\pi}\left(3\cos^2\theta+\cos^4\theta\right)\right)d\theta}{\int_{-\pi}^{\pi}\cos\theta d\theta+\int_{-\pi}^{\pi}(1+\cos^2\theta)d\theta}$$

$$= \frac{2a}{3}\frac{\int_{-\pi}^{\pi}\left(3\cos^2\theta+\cos^4\theta\right)d\theta}{\int_{-\pi}^{\pi}(1+\cos^2\theta)d\theta} = \frac{2a}{3}\frac{4\int_0^{\frac{\pi}{2}}\left(3\cos^2\theta+\cos^4\theta\right)d\theta}{4\int_0^{\frac{\pi}{2}}(1+\cos^2\theta)d\theta} = \frac{2a}{3}\frac{3\times\frac{1}{2}\times\frac{\pi}{2}+\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}}{\frac{\pi}{2}+\frac{1}{2}\times\frac{\pi}{2}} = a\frac{\frac{\pi}{2}+\frac{\pi}{8}}{\frac{\pi}{2}+\frac{\pi}{4}}$$

$$= a\frac{\left(\frac{5\pi}{8}\right)}{\left(\frac{3\pi}{4}\right)} = \frac{5a}{6}$$

Hence centre of gravity of the cardioid is at $\left(\frac{5a}{6},0\right)$

3. Find the moment of inertia of a hollow sphere about a diameter, its external and internal radii being 5 meters and 4 meters respectively.

Solution: Solution: let $\rho =$ a constant.

The moment of inertia about the diameter, i.e., X axis is, $I_x = \iiint_V \rho\left(y^2+z^2\right)dxdydz$.

Changing to spherical coordinates, we get,

$$I_x = \int_0^{2\pi}\int_0^{\pi}\int_4^5 \rho\left[(r\sin\theta\sin\phi)^2+\left(r\cos\theta\right)^2\right]r^2\sin\theta drd\theta d\phi$$

$$= \rho\int_0^{2\pi}\sin^2\phi d\phi\int_0^{\pi}\sin^3\theta d\theta\int_4^5 r^4dr+\rho\int_0^{2\pi}d\phi\int_0^{\pi}\cos^2\theta\sin\theta d\theta\int_4^5 r^4dr$$

$$= \frac{8\pi\rho}{15}\left(5^5-4^5\right) = 1120.5m$$

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